



Empowering Learners to Build Production - Ready
AI Agents

Applied Agentic AI for SWEs

Foundational Content

How to study Foundational Material?

Learner Resource Guide

Foundational Content – Learner Resource Guide

Applied Agentic AI for SWEs - Foundational Material -2026 - This resource collection is designed to support learners with **different levels of Python and GenAI familiarity**. To ensure a smooth learning experience without overwhelming learners, the foundational content is organized into **two clearly defined tracks**:

- **Before Class 1: Mandatory Python Resources(Required for the Course)** – the minimum required to successfully participate in the course and sufficiently understand how Large Language Models (LLMs) and AI agents are used and applied throughout the program.

Python itself is a broad subject, and learners who wish to go deeper can explore the detailed Python resources provided as optional content at their own pace.

- **Optional Deep-Dive Resources** – additional material for learners who want stronger conceptual depth and extended practice.

Together, these two tracks provide a structured path from **Python fundamentals to Agentic readiness**, while allowing learners to choose the depth based on their background and learning goals.

This guide explains how the foundational content is structured and how learners should use each resource collection effectively.

Python Setup & Learning Scope

To complete the mandatory Python content, you will need either **Google Colab** or a **local Python setup** on your computer. You may choose either option. We recommend installing both, as you will use them at different points throughout the course; however, for this starter kit, access to one is sufficient.

Detailed setup instructions for both options are available in the **next resource collection, *Python Starter Kit: Tools & Tutorials***.

For this course, the mandatory foundational Python content is covered in a **4-hour lecture**, which is sufficient to understand and work with LLMs and AI agents. The only requirement

before Class 1 is to complete the Mandatory Python Resources, and you have two weeks to do so.

Before class 1: Mandatory Python Resources (Required for the Course - Self-paced)

This resource collection contains the **minimum Python knowledge** required to successfully follow the course, complete hands-on exercises, build capstone projects, and effectively understand and work with Large Language Models (LLMs) and AI agents

Who should use this?

- All learners, regardless of prior Python experience
- Learners who want to quickly get up to speed with only what is required for the course

Topics Covered:

1. Introduction to Python, Data Types & Strings
2. Lists & Tuples
3. Sets & Dictionaries
4. Conditionals (If-Else Statements)
5. Loops
6. Functions
7. Modules & Packages
8. Pandas (Core Usage)

Note:

- Learners are expected to complete this mandatory python content before the first live session
- This content is sufficient to participate fully in the course

If you need any help, please reach out to the support team.

During the Course (Self-Paced): Optional Deep-Dive Resources

This resource collection is designed for learners who want to **go beyond the minimum requirements** and build a stronger conceptual foundation around Python, Generative AI, and text-based workflows.

Who should use this?

- Learners who are new to Python or GenAI concepts
- Learners who prefer deeper explanations and practise questions.
- Learners who want additional confidence before advanced Agentic AI topics

Topics Covered

1. Extended Python Foundations
 - a. Python fundamentals (refresher and deeper coverage)
 - i. Basic Programming Concepts with Python
 - ii. Variables, Data Types, Typecasting
 - iii. Operators - Arithmetic, Logical, Comparison
 - iv. if-else statement
 - v. Loops (for-while)
 - vi. Functions
 - vii. Lists: Common Methods (Built-in Functions)
 - viii. Strings: Common Methods (Built-in Functions)
 - ix. Tuples: Common Methods (Built-in Functions)
 - x. Sets: Common Methods (Built-in Functions)
 - xi. Dictionaries
 - xii. Python Modules
 - xiii. Functional Programming
 - xiv. Scope of Variables
 - xv. Call by Value and Call by Reference
 - xvi. Coding practise exercises
 - b. NumPy basics for numerical operations
 - c. Pandas for data manipulation and analysis
 - d. Matplotlib for basic data visualization
 - e. Exception handling
 - f. Object-Oriented Programming (OOPs) basics
 - g. File handling in Python
2. Evolution of Generative AI(to understand what GenAI is and how it evolved)
3. Basics of Prompt Engineering
 - a. Communicating effectively with AI tools
 - b. Types of prompts and prompt refinement
4. Basics of Text Processing
 - a. Fundamental concepts of Natural Language Processing (NLP)
 - b. Typical workflow for NLP tasks
 - c. Tokenization, part-of-speech tagging, named entity recognition, and parsing
 - d. Text representation
 - e. Bag of words and TF-IDF(Term Frequency-Inverse Document Frequency)
 - f. Word embeddings: word2vec

Mini Project: Hello Agent – CSV FAQ Agent

After one week, we will release a **small optional activity (mini project)** for learners to think through and attempt.

Note: We strongly encourage learners to attempt this mini project if you have time after completing the Mandatory Python Resources. This project will also be **revisited after Week 1** to reinforce key ideas and clarify agent behavior.

Note:

- This content is **optional** and not required to complete before week 1 class. Learners can go through them during the course at their pace.
- Use these resources if you want more conceptual clarity or additional practice.

How to Decide Which Resource to Follow:

- If you want to **move fast and focus on the course objectives**, stick to the **Mandatory Python Resources**
- If you want to **strengthen your Python foundation or revise concepts**, use the **Optional Deep-Dive Resources** alongside

Key Note for Learners:

Both resource collections are provided to support different learning styles and backgrounds. You do not need to complete both to succeed in the program. Focus on the mandatory resources first, and use the optional content as needed.

If you have questions about which resources to follow, please reach out to the Operations team for guidance.